



Responder Proximity System

A secure, in-vehicle view that shows an assigned first responder how close they are to the person who needs help - without exposing responder locations to the public.

Designed as a CAD/AVL and RapidSOS integration for emergency communications centers, agencies, and field responders.

THE DECISION

Build the view inside the dispatch chain

The concept is feasible, but it should not be a public “radar.” It should be an assignment-scoped proximity layer connected to the agency's computer-aided dispatch (CAD) and automatic vehicle location (AVL) systems.

01

Victim location

UNITED sends the emergency location, accuracy, timestamp, and incident context into an approved emergency-data pathway.

02

Responder location

The agency's AVL feed supplies the assigned vehicle's trusted GPS position. UNITED never publishes fleet locations.

03

Proximity view

The vehicle mobile data terminal (MDT) receives road distance, estimated time of arrival (ETA), direction, and data-freshness warnings.

Recommended product definition: “UNITED Responder Proximity” is a field-responder interface and integration service. It supplements existing CAD navigation; it does not dispatch units, replace 9-1-1, or expose responders on a consumer map.

Why this architecture fits public safety

- RapidSOS describes integrations that place emergency data inside existing CAD, call-handling, mapping, and field-responder workflows.
- National 911 guidance defines Next Generation 9-1-1 (NG911) as a secure, open-standards system that delivers emergency data to appropriate agencies and interoperates with first-responder networks.
- APCO's CAD procurement guidance expects unit GPS, calls, and officer locations to be plotted together so dispatchers can assign the closest appropriate unit.

The first ask

Request a RapidSOS technical discovery session, data-partner pathway, sandbox access, and introductions to one participating CAD/field-responder integration partner.

The first pilot

One agency, one dispatch center, one CAD vendor, and a small vehicle set in shadow mode before live operational use.

WHAT HAPPENS DURING AN ALERT

CAD remains the source of truth

UNITED contributes incident-side data. The emergency communications center validates and dispatches. The assigned vehicle sees only the information needed to reach the incident safely.

- 1. Person initiates UNITED.** The consumer experience may recognize the phrase “Stay Calm” or a manual emergency action, but confirmation and local emergency procedures govern transmission.
- 2. UNITED forms an incident package.** It includes a unique incident ID, location, horizontal accuracy, timestamp, movement state, callback/contact channel, and the minimum relevant context.
- 3. Approved intake reaches the emergency communications center.** RapidSOS Digital Alerts or another approved channel places the event into an operator workflow; local policy determines acceptance and handling.
- 4. CAD creates or links the call.** The dispatcher validates the incident, assigns the appropriate unit, and remains authoritative for status, priority, cancellation, and reassignment.
- 5. The assigned vehicle receives proximity guidance.** A CAD extension or MDT app combines the incident position with the vehicle’s AVL position and a routing engine.
- 6. The view closes automatically.** Access ends when the assignment is cancelled, transferred, cleared, or expires under agency policy.

Information shown to an assigned crew

- **Incident pin:** coordinates plus accuracy radius
- **Road distance:** route-based, not straight-line only
- **ETA:** identified as an estimate
- **Direction:** next turn or bearing

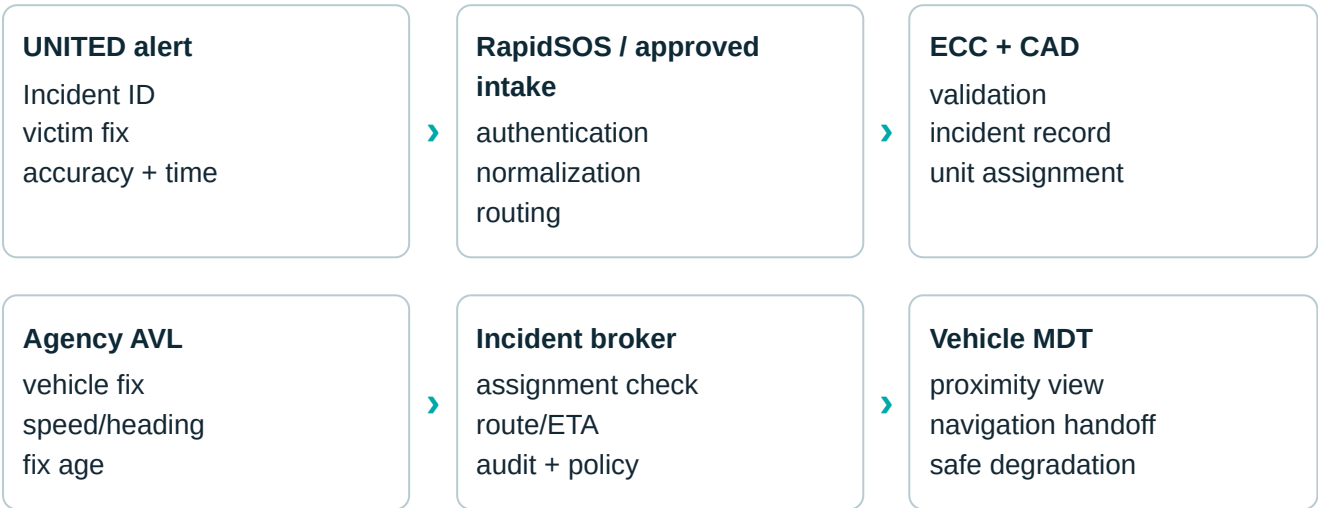
- **Freshness:** age of victim and AVL fixes
- **Movement:** stationary, moving, or unknown
- **Assignment:** dispatched, en route, arrived, cleared
- **Hazards:** only dispatcher-approved notes

Operational boundary: The vehicle view never self-dispatches, selects a unit, promises an arrival time to the victim, or bypasses the dispatcher. If any location is stale or uncertain, it says so plainly and falls back to the CAD location.

END-TO-END DESIGN

Join five trusted systems; create no public feed

The design uses agency and partner systems already responsible for emergency intake, dispatch, fleet location, and field delivery.



Proposed logical flow. Final interface placement depends on RapidSOS, the agency, and the selected CAD/MDT vendor.

Component responsibilities

Component	Owns	Must not own
UNITED	Alert-side data and consent	Dispatch decisions
RapidSOS / intake	Approved emergency-data delivery	Public responder tracking
ECC / CAD	Validation, assignment, status	Consumer app behavior
Agency AVL	Trusted vehicle position	Victim data
Incident broker	Authorization, correlation, metrics	Permanent fleet history
MDT client	Glanceable assigned-incident view	Independent dispatch

Preferred deployment choices

- CAD-native pane
- MDT deep link
- Agency SSO
- Managed device
- FirstNet-capable transport
- Offline-safe map cache

ILLUSTRATIVE SCREEN - NOT LIVE DATA

Call it proximity, not radar

“Radar” is a useful visual metaphor, but the system calculates proximity from GPS/AVL and routing data. The interface should say exactly what the data means.



ASSIGNED INCIDENT

Medical assistance

ROAD DISTANCE

1.8 mi illustrative

ESTIMATED ARRIVAL

4 min traffic-adjusted

VICTIM FIX

8 sec old ± 18 m

VEHICLE FIX

3 sec old

Dark diamond: assigned unit. Red point: incident. Rings: relative distance bands, not radio detection.

Moving mode

- Map, distance, ETA, next turn, alert age
- No scrolling narrative or manual data entry
- Large targets; optional spoken updates
- All nonessential detail locked while moving

Stopped mode

- Full incident notes and contact workflow
- Accuracy history and location-source detail
- Floor/entrance data when available
- Supervisor and dispatcher actions

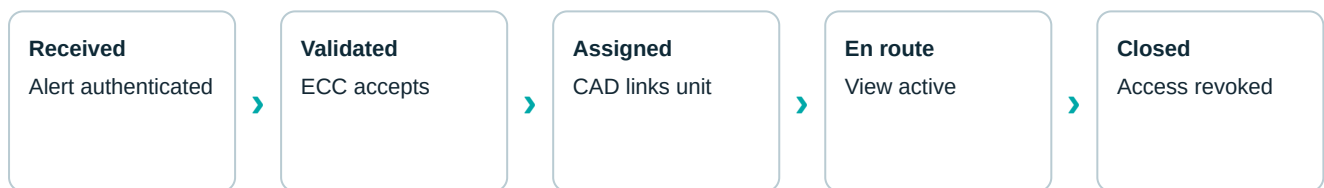
Safety rule: Set the destination automatically from CAD. The driver should not type an address or manage the incident while moving. A passenger or dispatcher handles deeper interaction.

MINIMUM TRUSTWORTHY EXCHANGE

Every location needs identity, accuracy, and age

Object	Required fields	Rule
Incident	incident_id, source_id, type, priority, state	CAD ID linked after acceptance
Victim fix	lat, lon, accuracy_m, observed_at, source	Never display an unaged point
Vehicle fix	unit_id, lat, lon, heading, observed_at	Agency AVL only
Assignment	incident_id, unit_id, status, assigned_at	Required before field access
Route result	road_distance, ETA, calculated_at, provider	Clearly marked as estimate
Audit event	actor, device, action, time, outcome	Append-only trail

Event lifecycle



Interoperability posture

- **Map local CAD fields to a neutral internal model.** Support vendor adapters rather than hard-coding one CAD.
- **Align incident exchange with public-safety standards.** APCO/NENA's Emergency Incident Data Document (EIDD) is an established XML/NIEM model; later NENA i3 material also defines a JSON-based Emergency Incident Data Object (EIDO).
- **Use common status semantics.** Normalize assigned, acknowledged, en route, arrived, cancelled, and cleared across vendors.
- **Prefer event delivery over repeated polling.** Signed webhooks or managed message queues reduce delay and provide auditable delivery.

Location policy: Show the best authorized fix, its source, its age, and its uncertainty. When fixes disagree, do not silently blend them; elevate the discrepancy to dispatch.

NON-NEGOTIABLE CONTROLS

Responder location stays inside the agency

The system processes sensitive victim and fleet location. Access must follow the assignment, not curiosity, proximity, or possession of a link.

Identity and access

- Agency single sign-on and multifactor authentication
- Managed, registered vehicle devices
- Role- and assignment-based access
- Short-lived tokens; automatic revocation
- No consumer access to vehicle coordinates

Data protection

- Encryption in transit and at rest
- Per-agency tenant separation
- No sensitive data in URLs or push previews
- Minimal local cache with remote wipe
- Configurable retention and legal hold

Operational resilience

- CAD remains authoritative
- Visible stale-data and outage states
- Last-known fix never shown as current
- Redundant transport where the agency permits
- Graceful fall back to radio/CAD workflow

Oversight

- Immutable access and change logs
- Supervisor review and export
- Incident-bound analytics, de-identified by default
- Security testing before production
- Breach, outage, and misuse response playbooks

Required governance decisions before live use

Decision	Owner	Evidence
Who may view victim location?	Agency	Role/assignment policy
How long are fixes retained?	Agency counsel	Retention schedule
What data is CJI or PHI?	Agency + counsel	Classification review
What happens on mismatch?	ECC operations	Written SOP
Who may share across agencies?	Jurisdiction leaders	Agreement + audit

CJIS obligations depend on the deployed data and agency use. Treat compliance as an agency-led determination, not a product label.

PHASED DELIVERY

Prove the workflow before anyone relies on it

PHASE 0 - DISCOVERY

Confirm the doors

Document one agency's CAD, AVL, MDT, mapping, identity, network, records, and operating procedures. Agree who owns every interface.

Exit: signed architecture and data-flow decision.

PHASE 1 - SANDBOX

Build with synthetic incidents

Connect a UNITED test event to a non-production broker and vehicle emulator. Implement identity, event state, route/ETA, freshness, and audit.

Exit: replayable end-to-end tests.

PHASE 2 - SHADOW MODE

Observe, do not guide

Mirror selected training or approved incidents. Compare the proposed view with CAD and radio outcomes; field users do not depend on it.

Exit: agency-approved safety and accuracy report.

PHASE 3 - CONTROLLED OPERATIONS

Limit scope tightly

Enable a small trained cohort, one incident class, and defined hours with dispatch oversight and an immediate kill switch.

Exit: formal operational acceptance.

Pilot team

RapidSOS technical partner

ECC operations lead

CAD/AVL vendor

Agency GIS

Security/CJIS officer

Field responder lead

UNITED engineering

Planning assumptions

- A single-jurisdiction pilot is the fastest credible path.
- Sandbox and shadow-mode access precede production credentials.
- The agency provides approved test incidents and fleet-location access.
- Schedule and cost are estimated only after interface discovery; no fixed dates or prices are asserted here.

Stop conditions: any unauthorized location exposure, assignment mismatch, unexplained stale fix, unsafe driver interaction, or inability to revoke access ends the pilot until corrected.

EVIDENCE BEFORE LAUNCH

Measure safety and integrity, not just screen speed

Test area	Pass evidence	Failure response
Incident correlation	No cross-incident or cross-unit display in the test set	Block release; trace ID mapping
Location freshness	Age and accuracy shown for every fix	Suppress guidance; fall back to CAD
Assignment control	Only assigned, authenticated units receive details	Revoke tokens; security review
Status synchronization	Cancel, reassign, arrive, and clear propagate correctly	Hide view; alert dispatch
Network loss	Offline/stale state is unmistakable; no false current fix	CAD/radio workflow continues
Driver usability	Core guidance readable at a glance; no moving-mode typing	Simplify interface and retest
Auditability	Every view and location update attributable and time-stamped	Reject pilot evidence
Resilience	Documented recovery and kill-switch exercise succeeds	Return to shadow mode

Metrics to agree with the pilot agency

Technical

- End-to-end alert latency distribution
- Victim and vehicle fix age
- Route/ETA calculation success
- Availability and recovery time
- Event delivery and correlation errors

Operational

- Dispatcher corrections required
- Wrong-entrance or wrong-pin events
- Field-user comprehension
- Driver-distraction observations
- Time from assignment to arrival

Evaluation principle: Do not claim improved response time from a small pilot alone. First demonstrate that the system is accurate, usable, secure, and does not disrupt dispatch. Any outcome claim needs an agency-approved study design and baseline.

Release checklist

- | | |
|---|--|
| ☐ Data-sharing agreement approved | ☐ Retention and records policies configured |
| ☐ Threat model and penetration test complete | ☐ Fallback and kill switch exercised |
| ☐ Dispatcher and responder training complete | ☐ Agency signs operational acceptance |

PROPOSED PARTNERSHIP CONVERSATION

Ask for the integration path, not fleet access

UNITED is seeking a route to deliver trusted incident-side data into public-safety workflows and pair it, under agency control, with the assigned unit's existing AVL data.

One-sentence proposal

Partner with UNITED, one emergency communications center, and one CAD/field-responder vendor to test an assignment-scoped in-vehicle view of victim proximity, road distance, ETA, accuracy, and data freshness.

Requested from RapidSOS

1. **Technical discovery:** identify the correct product and partner pathway for UNITED-generated emergency data.
2. **Sandbox:** provide non-production documentation, credentials, sample payloads, and test procedures under RapidSOS terms.
3. **Integration guidance:** confirm how incident IDs, location updates, cancellations, and field-responder delivery should be represented.
4. **Ecosystem introduction:** identify a willing CAD/MDT partner and candidate public-safety agency for a limited pilot.
5. **Certification path:** define security, reliability, support, and validation gates for any "RapidSOS Ready" or partner designation.

Questions for the meeting

- Which RapidSOS product receives an app-originated emergency event with updated location?
- Can that event be linked to a CAD incident and exposed to an assigned field unit through an existing integration?
- Which party should calculate road distance and ETA: CAD, field app, mapping partner, or UNITED?
- What data may be retained, and what must be deleted when the incident closes?
- What sandbox, test PSAP/ECC, and certification resources are available?
- Which security, uptime, support, and incident-response commitments are required?

Not requested: a public API showing responder locations; permission for UNITED to choose or dispatch units; production access before agency approval; or replacement of CAD, radio, 9-1-1 call handling, or responder navigation.

WHAT TO PREPARE

A credible pilot starts with six artifacts

1**Product demonstration**

Show UNITED creating a test incident, including the “Stay Calm” trigger path, confirmation gate, location, accuracy, timestamp, and cancellation behavior.

2**Data dictionary**

List every field UNITED can send, where it comes from, how often it updates, and how missing or stale values are represented.

3**Threat model**

Document spoofing, replay, account takeover, unauthorized tracking, wrong-incident linking, device loss, and denial-of-service controls.

4**Integration adapter**

Build the neutral incident model and a sandbox adapter. Keep CAD- and RapidSOS-specific code behind defined interfaces.

5**Vehicle prototype**

Deliver a managed-tablet prototype that consumes only synthetic or agency-approved test data and demonstrates stale-data behavior.

6**Pilot packet**

Include the concept of operations, architecture, test plan, privacy assessment, training outline, support plan, and stop conditions.

Sequence: demonstrate the alert and data first; agree the partner pathway second; build the field prototype only after the agency and CAD/MDT interfaces are known.

A FOCUSED FIRST CONVERSATION

Use 30 minutes to identify the real integration path

The goal of the first meeting is not approval to go live. It is to identify the right RapidSOS product team, interface, sandbox, CAD partner, and public-safety pilot owner.

Time	Topic	Outcome
0-5 min	Problem and demo	Shared use case
5-12 min	Proposed data flow	Correct product path
12-20 min	CAD/AVL and field delivery	Partner dependency map
20-26 min	Security and pilot controls	Gate expectations
26-30 min	Owners and next step	Named follow-up

Bring

- Two-minute UNITED demo
- One-page data dictionary
- This architecture diagram
- Named technical contact

Leave with

- Correct integration channel
- Sandbox owner and requirements
- CAD/field partner candidate
- Date and owner for next step

Recommended close: “Can we schedule a technical session with the RapidSOS team that handles app-originated emergency data and field-responder/CAD integrations, then identify one sandbox and one potential pilot partner?”

GROUNDING

Public sources informing this plan

1. **RapidSOS - Public Safety Software Partners.** Describes integration with CAD, dispatching, call handling, mapping, accurate positioning, logging, and systems of record. <https://rapidsos.com/public-safety-partners/>
2. **RapidSOS - Digital Alerts.** Describes standardized digital incident data delivered into RapidSOS UNITE from connected devices and systems. <https://rapidsos.com/unite-digital-alerts/>
3. **National 911 Program - NG911 Interstate Playbook, Chapter 2.** Defines NG911 as secure, IP-based, open-standards infrastructure supporting added data and first-responder interoperability; recommends phased implementation. https://www.911.gov/assets/National_911_Program_NG911_Interstate_Playbook_Chapter-2.pdf
4. **APCO sample RFP guidance hosted by 911.gov.** Includes CAD mapping and AVL requirements for plotting fleet units, incidents, and officer locations. <https://www.911.gov/assets/2020-APCO-RFP-Template-Guidance.pdf>
5. **APCO/NENA Emergency Incident Data Document.** Describes an industry-neutral, NIEM-conformant exchange for emergency incident information. <https://www.apcointl.org/standards/2-105-1-2017-nena-apco-ng9-1-1-emergency-incident-data-document-eidd/>
6. **FirstNet Authority - Situational Awareness Roadmap.** Identifies location services and mapping/GIS integration for people, vehicles, assets, incidents, traffic, and hazards as core public-safety needs. https://www.firstnet.gov/sites/default/files/FirstNet_Authority_Roadmap_Situational_Awareness.pdf
7. **FBI - CJIS Security Policy v5.9.5.** Provides controls relevant to mobile devices, authentication, encryption, and managed access. Agencies must determine current applicable requirements. https://le.fbi.gov/file-repository/cjis_security_policy_v5-9-5_20240709.pdf
8. **U.S. Department of Transportation - driver-distraction guidance.** Highlights limiting manual entry and visual-manual tasks in moving vehicles. <https://www.transportation.gov/briefing-room/us-department-transportation-proposes-%E2%80%98distraction%E2%80%99-guidelines-automakers>

Important limitation: This is a concept and pilot proposal, not an assertion that RapidSOS exposes responder AVL data or has approved UNITED. The production design depends on RapidSOS, an emergency communications center, the agency, and its CAD/AVL/MDT vendors.

Terminology: ECC = emergency communications center; PSAP = public safety answering point; CAD = computer-aided dispatch; AVL = automatic vehicle location; MDT = mobile data terminal; GIS = geographic information system.